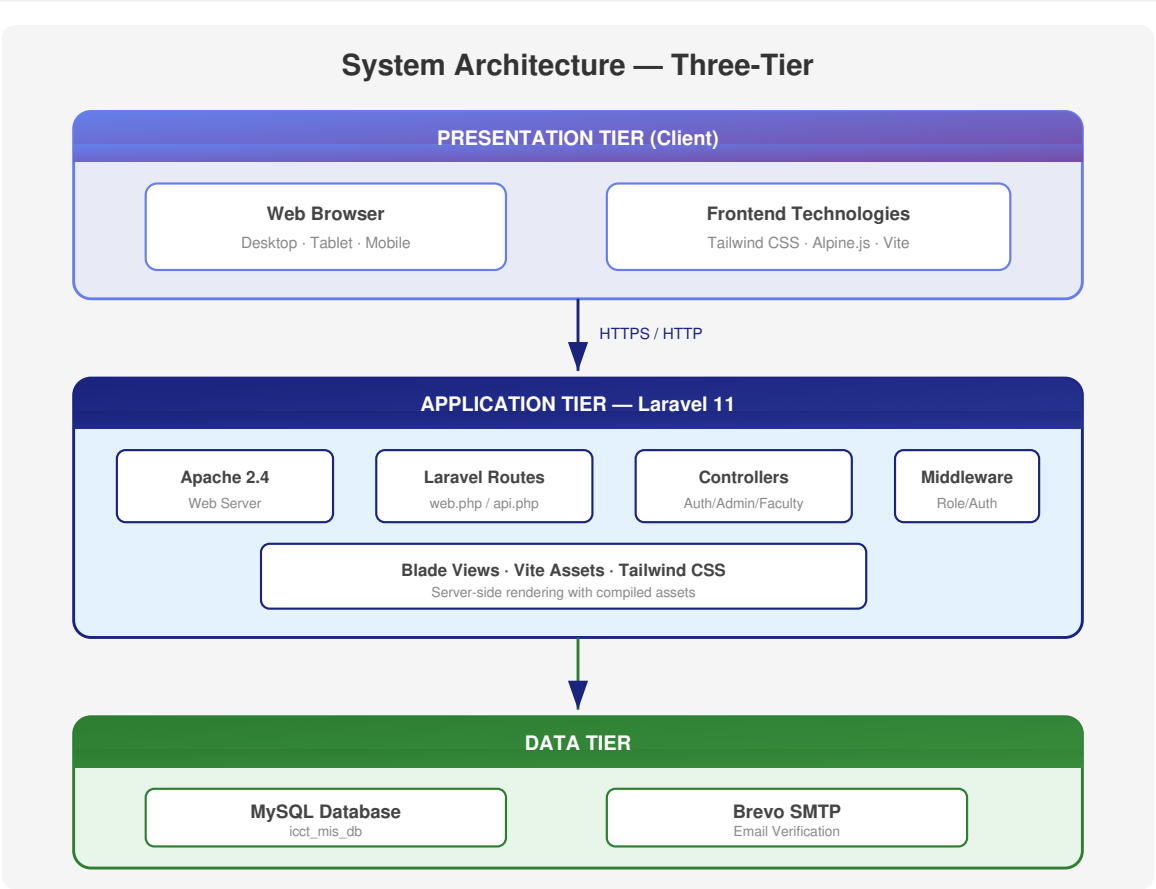


ICCT Cainta Campus — MIS Resource Management System

System Diagrams and Descriptions

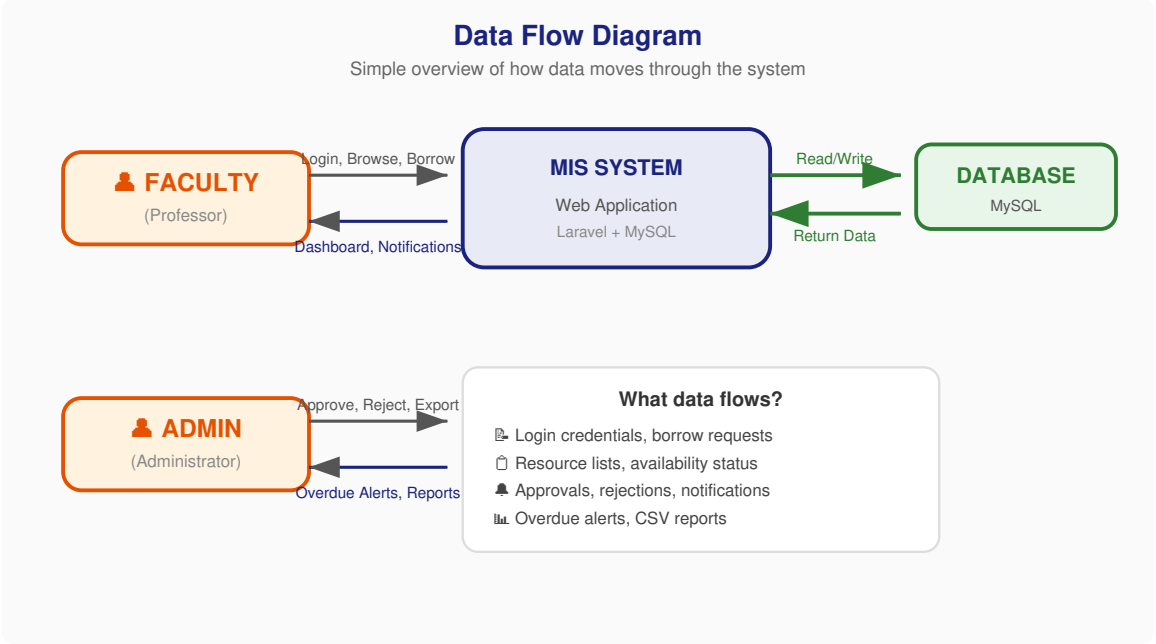
1. System Architecture Diagram

Description: The system follows a **three-tier architecture**. The **Presentation Tier** consists of the client web browser running Tailwind CSS and Alpine.js for responsive design on desktop, tablet, and mobile devices. The **Application Tier** runs Laravel 11 on Apache with MVC architecture — handling routing, business logic, authentication, and server-side rendering. The **Data Tier** uses MySQL database for storing all system data and Brevo SMTP for email verification services. Communication between tiers uses HTTPS for security.



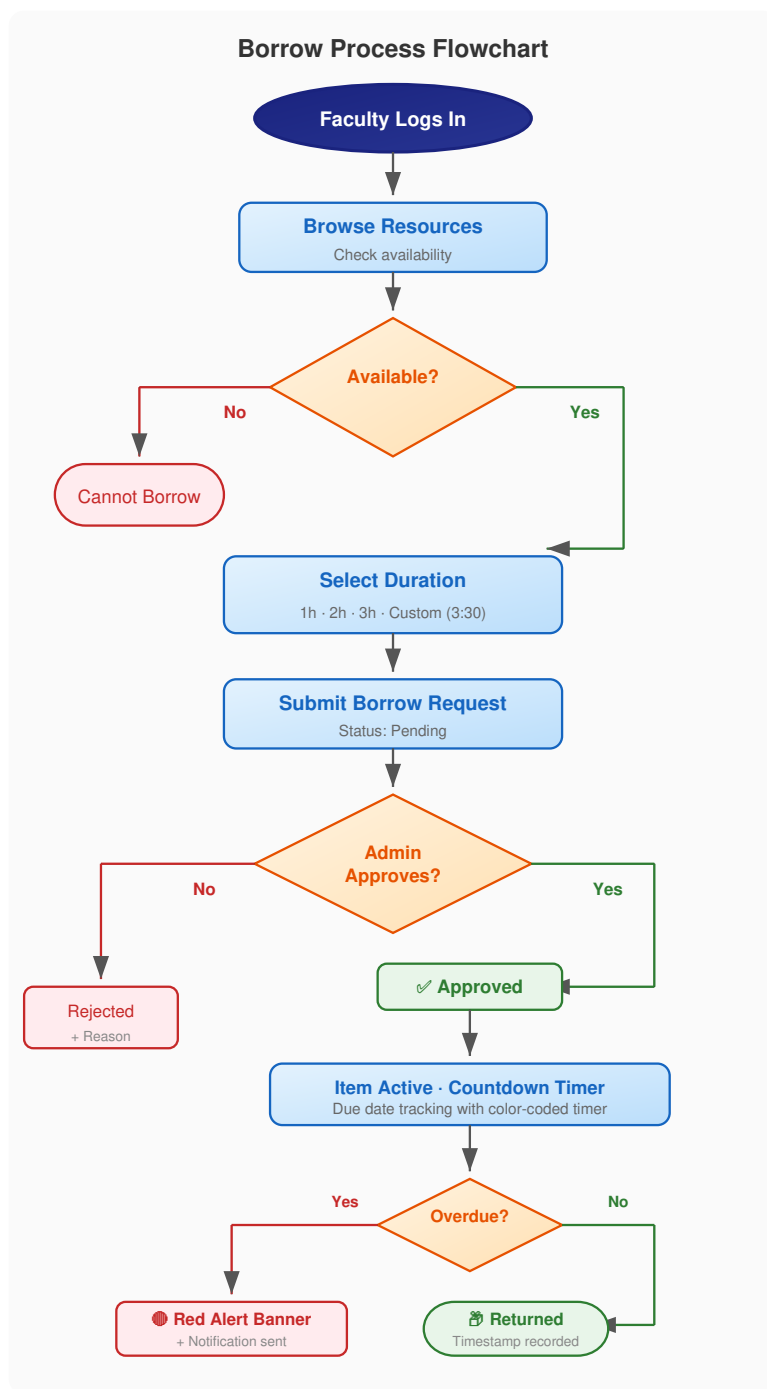
2. Data Flow Diagram (DFD Level 0)

Description: The DFD shows how data flows between external entities (**Faculty User** and **Administrator**) and the **MIS System**. The system is composed of six modules: (1) Authentication handles login and registration, (2) Resources manages resource browsing and availability display, (3) Borrow Manager processes borrowing requests with duration selection, (4) Approval enables administrators to approve/reject requests and mark items returned, (5) Notifications sends in-app alerts, and (6) Reports generates CSV exports. All modules read from and write to the **MySQL Database**. The **Brevo Email Service** is used for email verification.



3. Borrow Process Flowchart

Description: The flowchart illustrates the complete borrowing process from login to return. A faculty member logs in, browses available resources, and selects a borrowing duration (1 hour, 2 hours, 3 hours, or custom). If the resource is available, the system creates a pending request. The administrator reviews and either approves or rejects the request with a reason. Once approved, the item becomes active with a real-time countdown timer showing remaining time. The system automatically detects overdue items and displays a red alert banner. The process ends when the item is returned and marked as completed.

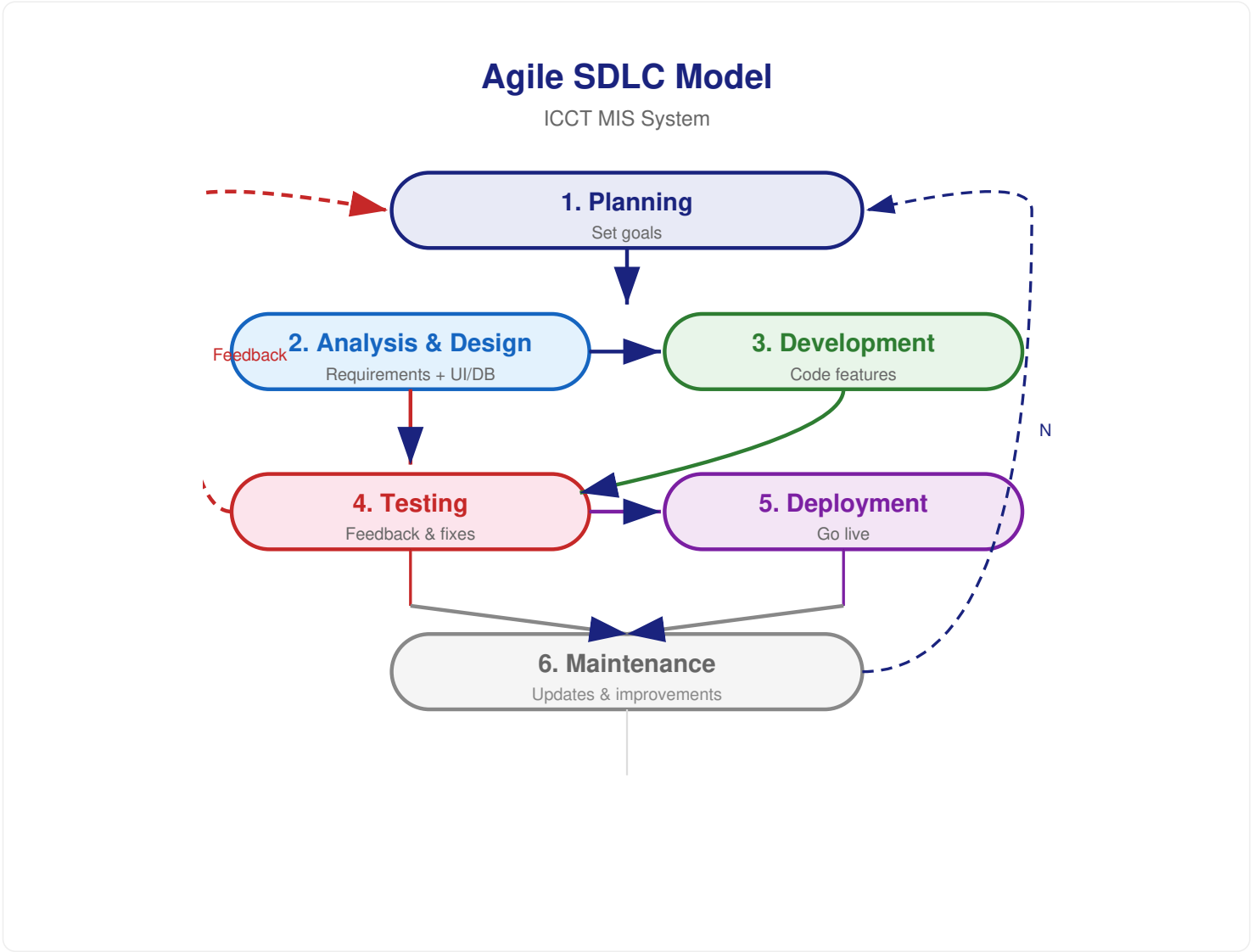


4. IPO Chart

IPO Chart — Input, Process, Output
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INPUT	PROCESS	OUTPUT
Login Credentials Email & Password	Authentication Validate credentials, check role	Dashboard Access Role-based dashboard view
Borrow Request Item, quantity, duration (1h/2h/3h)	Create Pending Request Save to DB, notify admin	Pending Record Waiting for admin approval
Admin Approval Approve or Reject + reason	Process Decision Update status, create notification	✔ Approved / ✕ Rejected Active borrow or rejected notice
Time Passes Due date arrives / passes	Cron: Check due_at < now() Detect overdue, send alert	🕒 Overdue Alert Red banner + notification
Admin marks Returned Click "Mark Returned"	Update Status Set returned_at, change status	📦 Item Returned Transaction complete
Export Request Admin clicks "Export CSV"	Generate CSV Format all records as CSV	📄 CSV File Downloadable report

Description: The system followed an **Agile/Iterative SDLC** approach. Each phase (Planning, Analysis, Design, Development, Testing) was completed in cycles, with continuous feedback from stakeholders driving improvements. Deployment was done after each stable iteration, and maintenance continues based on user feedback.



Appendix: Implemented Features Summary

#	Feature	Description	User
1	Authentication	Login/Register with email verification via Brevo, password recovery	All
2	Role-Based Access	Admin and Faculty roles with different dashboards and permissions	All
3	Resource Management	CRUD operations, auto-calculated availability, responsive views	Admin
4	Borrow Request	Digital submission with duration select (1h/2h/3h/custom HH:MM)	Faculty
5	Approval Workflow	Approve/Reject with reason, auto-notification, mark returned	Admin
6	Dashboard	Welcome overview, stats cards, overdue alert banner, weekly chart	All
7	Notifications	In-app with red badge, delete/clear, mark all read	All
8	Overdue Monitoring	Cron job, color-coded countdown, red alert banner with item details	All
9	Borrow History	Filter tabs (All/Pending/Approved/Borrowed/Returned/Rejected)	Faculty
10	CSV Export	Downloadable report of all borrow records	Admin
11	Mobile Responsive	Phone (cards), Tablet (compact), Laptop (full) layouts	All
12	Overdue Alert Banner	Red banner on dashboard listing overdue items with due dates	Faculty